User Manual

Mobile Controller System

This asset has been designed for competitive mobile games, such as mobile MOBA or battle royale type.
The behavior of this controller is exactly the same as a successful game that existed on the market.

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Features

- Strong scripting API
- Unrestricted modification
- Smooth and accurate mobile controller system
- Mobile have same behavior as editor
- All textures are easy to replace
- Clean, easy to understand C# code
- Allow unrestricted modification
- Based on Unity UI

Quick Start : Integration

1. Drag and drop “Mobile Controller” prefab into your scene.

2. This asset requires the EventSystem to work.
If not already exist, add EventSystem to the scene by Right Click on the scene’s hierarchy > UI > Event System

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Class UniversalButton

This is the base class of all buttons in this asset, Analog Stick also derived from this script.
Handle various touch inputs for button activation.

Class properties

Variable	Type	Remark
scaler	CanvasScaler	Reference of canvas scaler.
isAimable	bool	True = Can aim and activate. False = Activate on pressed.
btn	RectTransform	This button reference.
aimer	RectTransform	Aimer UI reference.
pointer	RectTransform	Pointer UI reference.
skillCanceller	RectTransform	Skill Cancel button reference.
hasText	bool	True if this button has a Text UI.
text	TextMeshProUGUI	Text UI of this button. Use with SetText function.
img	Image	Image component of this button.
state	ButtonState	Current state of this button.
isActive	bool	True = this button will accept touch input.
btnRadius	float	Radius of this button in pixel.
aimerRadius	float	Radius of aimer in pixels.
isManualAimOverride	bool	True when press and drag for aiming.
isFingerDown	bool	True when any finger is pressing this button.
isPointerUpOutOfBound	bool	True when pressed finger drag out of button radius.
initialFingerPosition	Vector3	First point when pressed on this button.
fingerId	int	System's finger ID that pressed on this button.
fingerPosition	Vector3	Current touch position on screen, pixel unit.
direction	Vector3	Aim direction in XY plane, value range of [-1,1]]
directionXZ	Vector3	Adjusted direction in XZ plane for easy use.
rawDir	Vector3	Raw direction value in pixel unit.
cancellerRadius	float	Cancel button radius in pixel.
horizontal	float	Return direction.x
vertical	float	Return direction.y
colorActive	Color	This button color when isActive = True.
colorInactive	Color	This button color when isActive = False;
colorPressed	Color	This button color when Pressed.

**The highlighted field is probably your frequently used parameters.*

Class events. Kindly look at the handleable events from the table below.

Events	Remark
onPointerDown (int)	Int btnIndex Fired on touch.
onBeginDrag (int)	Int btnIndex After pressed, fire when start moving finger from the initial touch position.
onDrag (int)	Int btnIndex After onBeginDrag. Fired every frame the finger moved.
onPointerUp (int)	Int btnIndex Fire on release touched finger.
onEndDrag (int)	Int btnIndex After onBeginDrag. After onPointerUp. Fired on release of the finger.
onActivateSkill (int)	Int btnIndex Fired on Pressed > Release Pressed > Drag > Release Use this even along with btnIndex and aiming direction parameters to get activation position.
onCancelSkill (int)	Int btnIndex Fired on Pressed > Drag > Release on Cancel button.

Use btnIndex parameter to identify which button own the event.

Public function.

Function	Remark
SetText (string)	Set the Text UI of this button. Use this for setting skill name or set skill cooldown value.

Class AnalogStick

Extended from UniversalButton class
- Output direction and amount
- Reposition itself based on initial touch input
- Will not reposition if initial touch input is in close proximity of aiming circle
- Aiming circle will not go out of screen boundary

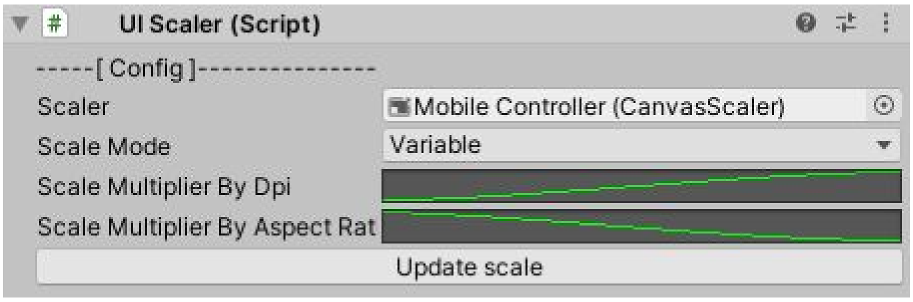
Events	Remark
onPointerDown (int)	Int btnIndex Fired on touch.
onBeginDrag (int)	Int btnIndex After pressed, fire when start moving finger from the initial touch position.
onDrag (int)	Int btnIndex After onBeginDrag. Fired every frame the finger moved. Use this to get real time direction value.
onPointerUp (int)	Int btnIndex Fire on release touched finger.
onEndDrag (int)	Int btnIndex After onBeginDrag. After onPointerUp. Fired on release of the finger.
onCancelSkill (int)	Int btnIndex Fired on Pressed > Drag > Release on Cancel button.

Use btnIndex parameter to identify which button own the event.

Variable	Type	Remark
dpadInner	RectTransform	Inner radius of AnalogStick if user initial touch input lands exactly on dpadInner radius. AnalogStick will position itself exactly on the initial touch position. This makes AnalogStick spawn with zero diction value as intended. If the user's initial touch lands outside of dpadInner but still within the dpadOuter area, AnalogStick will spawn with a direction value for the user wanting to move the character in that direction as intended.
dpadOuter	RectTransform	Area that can accept touch input for AnalogStick.
directionalPointer	RectTransform	Arrow UI that points to the aiming direction of AnalogStick.
dpadCosmetic	RectTransform	Appearance UI for AnalogStick.
innerRadius	float	dpadInner radius in pixel.
pointerRadius	float	directionPoint radius in pixel.
dpadInner	RectTransform	Inner radius of AnalogStick if user initial touch input lands exactly on dpadInner radius. AnalogStick will position itself exactly on the initial touch position. This makes AnalogStick spawn with zero direction value as intended. If the user's initial touch lands outside of dpadInner but still within the dpadOuter area, AnalogStick will spawn with a direction value for the user wanting to move the character in that direction as intended.
dpadOuter	RectTransform	Area that can accept touch input for AnalogStick.

Custom UI Scaler

This asset uses a custom UI scaler script.



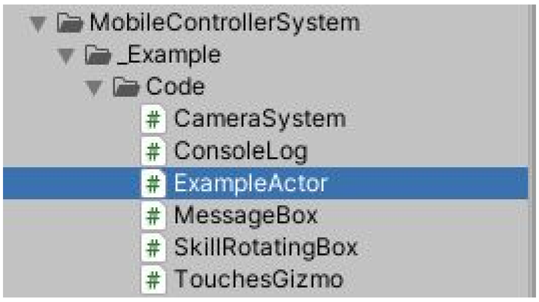
Usage : Adjust the 2 curves and test the result on actual devices.

The custom UI Scaler solved the problem for UI size across multiple screen sizes out of the box.
With our UI Scaler, developers could specify UI size for every device screen size and aspect ratio by graph UI and custom algorithm not manual labor.
And also allow developers to update their scaling parameters through API and in real-time or allow custom algorithms to be implemented.

This could be further developed into allowing users to make changes to their UI size and scale for serious pro gamer.

Example

Kindly take a look at ExampleActor.cs for more code examples.



This script shows how to handle various button events/parameters.

If you have any questions kindly reach out to us via email : support@suriyun.com